Capstone Final Report Phase-VII

FORMULA 1 Race Intelligence Dashboard - Project Report

Student Name: Pranay Patil	**Enrollment No:** EN25EL3010049	**Maximum Marks:** 10 Marks
College / University: Medi-Caps University	**College Email:** en25el3010049@medicaps.ac.in	**Date:** October 2026
Project Title: FORMULA 1 Race Intelligence Dashboard Analysis Report (2010 – 2026)		**Domain:** Data Analytics with Tableau

1. Executive Summary & Introduction

Formula 1 represents the pinnacle of international motorsport, where competitive differentiation hinges upon split-second strategy, aerodynamic efficiency, and powertrain reliability. Over 17 seasons spanning 2010 to 2026, comprising 6,464 race entries across 34 circuits and 4 regulatory eras, this capstone analytics project delivers a multi-dimensional intelligence dashboard engineered in Tableau. The solution empowers Team Strategist Rajan, Broadcast Analyst Divya, and FIA Compliance Officer Arvind with data-driven decision tools.

Student Submission Metadata:

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2. Key Business Scenarios & Analytical Findings

Scenario & Stakeholder	Key Analytical Finding	Strategic / Operational Action
Scenario 1: Team Strategist (Mr. Arnold / Rajan)	Mid-tier teams average 114 pts vs. frontrunner Red Bull's 612 pts (498-pt deficit). Soft tyre degradation averages +0.082s/lap, reaching cliff at Lap 18 on balanced circuits.	Adopt 1-stop Soft-to-Hard on high downforce street circuits; mandate 2-stop Medium-Hard-Soft on high-energy circuits to overcome the 22.4s pit penalty.
Scenario 2: Broadcast Media (Ms. Lisa / Divya)	Lewis Hamilton (4,825.5 pts, 86 wins) and Max Verstappen (3,110.5 pts, 63 wins) lead all-time standings. Italy (28 GPs) and UK (18 GPs) dominate race hosting.	Deploy automated Top-10 leaderboards and podium ratio infographics during pre-race telecasts to drive viewer engagement.
Scenario 3: FIA Compliance (Mr. Gilchrist / Arvind)	Grid reliability surged from 81.8% in 2010 to 92.4% in 2026. Mechanical engine blowouts fell by 48% following power unit component quotas.	Validates that aerodynamic regulations and powertrain standardization significantly enhanced vehicle durability and track safety.

3. Conclusion & Recommendations

The integration of 9 multi-dimensional worksheets into an interactive 3x3 Tableau telemetry dashboard solves data siloing across team strategy, live media broadcasting, and FIA governance. As Formula 1 transitions into the 2026 Active Aerodynamics and 100% Sustainable Fuel era, this analytics framework provides the foundation for real-time race intelligence and powertrain life-cycle monitoring.